Progressive exercises

E1 · Direct application. Consider the sequence (u_n) defined, for every natural number n by $u_n = 5 - 3n$. Calculate u_4, then the sum $\sum_{k=0}^{20} u_k$.

E2 · Application and justification. Next (v_n) is defined by $v_0 = 3$ and, for every natural number n, $v_{n+1} = 2v_n$. Express v_n as a function of n, then calculate $\sum_{k=0}^{n} v_k$.

E3 · Method to organise. Prove by induction that, for every natural number n, $3^n \geq 2n + 1$.

E4 · Choice of strategy. Consider the sequence (u_n) defined on $\mathbb{N}$ by $u_n = \frac{n+1}{n+2}$. Prove that it is increasing and give a bracketing valid for all its terms.

E5 · Synthesis or proof. Prove by induction that, for every natural number n, $8^n - 1$ is divisible by 7. Congruences will not be used.